



Luxottica Glass

Google ties up with luxury brand Luxottica to release new Glass models
<http://dgit.in/LuxGlass>



Chrome Now

Google Now notifications will be available on Chrome desktop browser
<http://dgit.in/ChromeNow>

Work @ tech

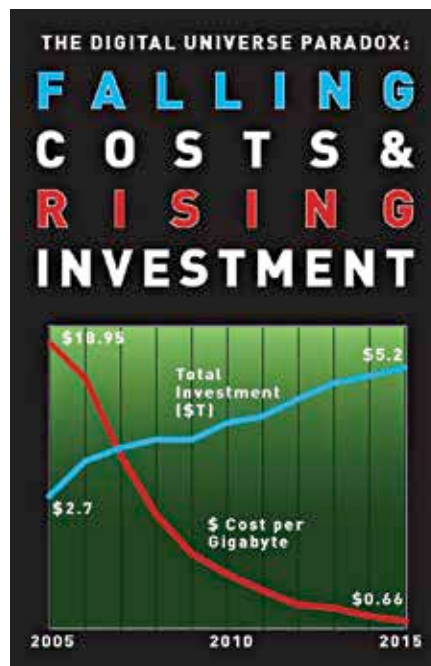
ising it requires a different approach. For example, people are complicated; they have friends, a transaction history, tweets and a plethora of other social media components, which form a rich complex structure that you need to understand and analyse. The software that you use should allow you to quickly inject the data into the data store, and allow it to have an analytic framework around it. It should be able to quickly retrieve data that's important to you.

The evolution of data has led to the development of new database platforms such as Hadoop and NoSQL, which meets the above requirements. These platforms are designed to store large volumes of complex data, which don't nicely fit into tables. Presently there's high demand for online and mobile data storage, which has also affected the hardware being employed to keep large volumes of data online rather than moving them to archives.

Why Database Administrator?

A Database Administrator, or DBA, manages the database in a way the industry demands. To do so, he uses his programming and administration skills. We mean that the database administrator is responsible for the installation, configuration, upgrade, maintenance and security of an organisation's database. The overall goal is to keep the servers up and running at all times and provide the users with access to the required information when needed. The DBA must protect the database from damage – accidental or intentional – and minimise the chances of data loss. While this is the traditional role of a DBA in an organisation, modern times call for more. Now just getting the required information isn't enough, the database admin must find a way to efficiently exploit the information. "In a time where everything in this world relies on data, and that too digital data, we can't compromise on accuracy and timeliness, when the slightest lack of precision or delay by microseconds may cause a company to lose millions", says Avinash Mathur, an Oracle DBA for a reputed IT consulting firm.

"The role of a DBA is evolving", explains Kamal. Traditionally, it was about managing a stable, relatively slow moving technology, which is the relational database.



Digital Universal Paradox (IDC Digital Universe Study 2011)

Today, a DBA has to deal with many aspects and not just administration. He continued, "Their role is moving towards making decisions on how they can manage more complex systems built on Big Data technologies, that encompasses things like Hadoop, NoSQL-based MongoDB, and potentially existing legacy relational data stores."

Competencies required for DBA

If you're considering a career in databases, you must be skilled in Structured Query Language, or SQL, as it's more commonly known. SQL is a programming language designed especially to manage and access data in a relational database management system (RDBMS). What is an RDBMS, you ask? A relational database is the most common database model that we've all used at some point in our lives. The spreadsheets that you make are a befitting example of a relational database. If you haven't heard of it before, don't worry. RDBMS is just a fancy term geeks use to refer to a collection of tables, in which data is organised in rows and columns. A database administrator is expected to know about popular relational database systems such as Oracle, MySQL, Microsoft SQL Server, PostgreSQL and IBM DB2.

Relational databases are the most common and convenient way of organising data. However, there's an increasing demand for other tools to manage data, since it isn't always feasible to organise data into rows and columns. Oracle DBA, Avinash tells us that there are various other tools available in the market for Big Data analytics:

- Hadoop
- NoSQL Database
- Pentaho Business Analytics
- Karmasphere Studio and Analyst
- Talend Open Studio
- Skytree Server

Apart from these, a database administrator should have a good understanding of the operating system and network architecture (client/server, intranet/internet, etc) he'll be working on. Companies want to hire "developers who have an understanding of the different data storage techniques and the type of technologies that are used to build applications", according to Kamal.

Everyday responsibilities

Let's talk in detail about the job responsibilities of a database administrator. This will help you get a good picture of what sort of work your duty will entail.

- Database administrators decide who gets access to which part of the database by assigning privileges to the end users, specific to their needs. He implements authentication and authorisation mechanisms to establish security of the database.
- He allocates the storage for the database in co-ordination with the IT department of the organisation. Both, present and future needs of the company are kept in mind when planning the storage capacity of the database.
- Database administrators should be well aware of the functioning of the database management system, and tackle all the technical problems that may arise.
- He writes and continually updates the documentation for the database. A database with an inconsistent or obsolete documentation is difficult to maintain – especially for people who didn't originally design it and will be working on it in the future.